

Experiment Trials

Trial A: Map-Reduce Test

The screenshot displays the Threat Hunter interface. On the left is a settings sidebar with sections for Model Selection (Manager: llama3.8b, Researcher: deepseek-r1.7b, Editor: qwen2.5.7b), Network (Interface: Wi-Fi, Packets: 15), Session ID (session_202620), and Live Capture (Interface: Wi-Fi, CAPTURE NOW button). The main panel shows a query: "Identify the packets from the live traffic capture for IP addresses that may be doing suspicious activities, then give me a status report on them." Below this, the "INTERFACE: Wi-Fi" section shows "PACKETS: 15" and a list of 9 packets. The "editor" section shows an "Analyst Feedback" message and a list of suspicious activities. The "manager" section shows a "Commander Suspicious Activity Report" with a list of suspicious activities. The "interrupt" button is at the bottom.

Settings

TShark ready

Model Selection

Manager: llama3.8b

Researcher: deepseek-r1.7b

Editor: qwen2.5.7b

Network

Interface: Wi-Fi

Packets: 15

Session ID: session_202620

Live Capture

Interface: Wi-Fi

CAPTURE NOW

Last:

Identify the packets from the live traffic capture for IP addresses that may be doing suspicious activities, then give me a status report on them.

researcher

Threat Hunter LIVE TRAFFIC CAPTURE:

INTERFACE: Wi-Fi PACKETS: 15

- 192.168.1.207 → 142.251.151.119 Protocol: UDP
- 142.251.151.119 → 192.168.1.207 Protocol: UDP
- 7 → 7 Protocol: ARP
- 7 → 7 Protocol: ARP
- 192.168.1.207 → 172.253.122.188 Protocol: TCP
- 172.253.122.188 → 192.168.1.207 Protocol: TCP
- 192.168.1.207 → 142.251.151.119 Protocol: UDP
- 192.168.1.207 → 142.251.151.119 Protocol: UDP
- 192.1

3.65s

editor

Analyst Feedback: The request is clear but could benefit from some additional details to ensure accuracy and completeness. For example:

- What specific types of suspicious activities are you looking for (e.g.

27.2s

manager

Commander Suspicious Activity Report

After analyzing the live traffic capture, I have identified several packets that may be indicative of suspicious activities:

- Packet 1-2: The exchange between IP addresses 192.168.1.207 and 142.251.151.119 appears to be a potential command and control (C2) communication, as it uses UDP protocol. This type of communication is often used by malware or attackers to send commands or receive data.
- Packet 5-6: The TCP connection between IP addresses 192.168.

117.75s

interrupt

Ask about cybersecurity (e.g., "Show live traffic", "Check IP 185.130.5.253")...

Latency:

Researcher-3.65s

Editor-27.2s

Manager-117.75

In the trial I asked the assistant to identify packets from the live capture results with IP addresses that may be doing suspicious activity and to give a status report on them. Upon observation, although running in parallel, the researcher and the editor nodes did not finish at the same time. The researcher node finished the fastest with 3.65s latency while the editor completed its task with a latency of 27.2s. What the reducer does is take the outputs, which are the IP packets and the editor deals with the context of the user query, in order to combine them into a final answer or report.

Trial B:Human-in-the-Loop Test

The screenshot displays a Human-in-the-Loop (HIL) testing interface. On the left, a control panel includes dropdown menus for 'llama3.8b' (Model), 'Researcher' (Role), 'deepseek-r1.7b' (Model), and 'Editor' (Role), with 'qwen2.5.7b' selected. Below these are sections for 'Network Interface' (set to 'Wi-Fi'), 'Packets' (a progress bar at 10), 'Session ID' (session_232424), and 'Live Capture' (Interface: Wi-Fi, with a 'CAPTURE NOW' button). A 'Last:' section shows a list of network packets: 1. 142.250.188.10 → 192.168.1.207, Protocol: UDP; 2. 192.168.1.207 → 142.250.188.10, Protocol: UDP; 3. ? → ?. On the right, a task list shows the progression of the analysis:

- Task 1: Explain the pros and cons of using a Null Scan on a windows machine. The response discusses early threat detection and system integrity maintenance.
- Task 2: Draft a concise, detailed summary of the live capture results. The response provides a summary of 10 packets, identifying two distinct communication flows (Flow 1 and Flow 2) and noting the use of TCP and TLSv1 protocols.
- Task 3: Packets 5-10 are not mentioned. Rewrite the detailed summary to include all the packets from the live capture results. The response is pending.

At the bottom right, a status bar shows 'researcher' and 'Threat Hunter' roles, along with a 'LIVE TRAFFIC CAPTURE' button.

The user provided human-in the loop feedback stating packets 5-10 were not mentioned and asked the manager node to reroute the task back to the researcher(threat hunter) and the editor(analyst) node. After receiving revised data from Editor and researcher nodes, it generated a comprehensive response which produced 1-8 packets due to it's output constraints. Below is the screenshot.

 Packets 5-10 are not mentioned. Rewrite the detailed summary to include all the packets from the live capture results

 researcher

 Threat Hunter  LIVE TRAFFIC CAPTURE:

INTERFACE: Wi-Fi PACKETS: 10

1. 192.168.1.207 → 142.250.188.10 Protocol: UDP
2. 142.250.188.10 → 192.168.1.207 Protocol: UDP
3. 13.69.239.74 → 192.168.1.207 Protocol: TCP
4. 192.168.1.207 → 13.69.239.74 Protocol: TCP
5. 192.168.1.207 → 142.251.167.189 Protocol: UDP
6. 142.251.167.189 → 192.168.1.207 Protocol: UDP
7. 192.168.1.207 → 162.159.130.233 Protocol: SSL
8. 192.168.1.207 →

 3.28s

 editor

  Analyst APPROVED

If you need any specific guidance on how to rewrite the summary, please let me know!

 8.76s

 manager

  Commander Here is the rewritten detailed summary to include all the packets from the live capture results:
Analysis:  LIVE TRAFFIC CAPTURE:

INTERFACE: Wi-Fi PACKETS: 10

1. 192.168.1.207 → 142.250.188.10 Protocol: UDP
2. 142.250.188.10 → 192.168.1.207 Protocol: UDP
3. 13.69.239.74 → 192.168.1.207 Protocol: TCP
4. 192.168.1.207 → 13.69.239.74 Protocol: TCP
5. 192.168.1.207 → 142.251.167.189 Protocol: UDP
6. 142.251.167.189 →

 122.03s

 interrupt

Trial C: Quantization Test

Q8 response from Assistant

The screenshot displays the 'Cybersecurity Incident Response Agent' interface. On the left is a blue sidebar with sections: 'Model Selection' (Manager: llama3:8b-instruct-q8_0, Researcher: deepseek-r1:7b, Editor: qwen2.5:7b), 'Network' (Interface: Wi-Fi, Packets: 10), 'Session ID' (session_004322), 'Live Capture' (Interface: Wi-Fi, CAPTURE NOW button), and 'Metrics' (Editor: 28.48s, Researcher: 191.1s, Manager: 130.52s). The main panel shows a chat log for the prompt 'Explain the pros and cons of using a Null Scan on a windows machine.' The chat includes an 'editor' message, a 'researcher' message titled 'Threat Hunter Null Scan on Windows Machines: Pros and Cons' with three pros listed, and a 'manager' message titled 'Commander Final Response' with the same three pros listed. The interface is partially cut off at the bottom.

Latency:

Editor -28.48s

Researcher-191.1s

Manager-130.52s

Note: final response does not complete.

Final Response:

This screenshot shows the same interface as the previous one, but the 'Final Response' from the manager is now visible. It repeats the three pros: 'No Prior Knowledge Required', 'Efficient Execution', and 'Stealthy Nature'. It also lists one con: 'Inaccurate Results: Null Scans can produce false posit'. The sidebar metrics remain the same.

Versus

Q2 response from Assistant

The screenshot displays the 'Cybersecurity Incident Response Agent' interface. On the left is a sidebar with controls: 'Model Selection' (Manager: llama3.8b-instruct-q2_K, Researcher: deepseek-r1.7b, Editor: qwen2.5.7b), 'Network' (Interface: Wi-Fi, Packets: 10), 'Live Capture' (Interface: Wi-Fi, CAPTURE NOW button), and 'Metrics' (Editor: 28.34s, Researcher: 164.58s, Manager: 65.96s). The main panel shows a chat history with a red prompt 'Explain the pros and cons of using a Null Scan on a windows machine'. The chat log includes an 'Analyst Feedback' message, a 'Threat Hunter' message, and a 'Commander' message. The 'Commander' message contains the final response: 'The Pros and Cons of Using a Null Scan on a Windows Machine'. The response lists three pros: Efficiency, Smooth User Experience, and Early Threat Detection. The Manager's response is partially visible at the bottom.

Cybersecurity Incident Response Agent

Explain the pros and cons of using a Null Scan on a windows machine

Analyst Feedback: The response seems to be missing. Could you please provide the response so I can verify its appropriateness?

Threat Hunter A Null Scan on a Windows machine offers several benefits, such as efficiency in identifying potential vulnerabilities without executing harmful code, providing a smooth user experience, detecting hidden risks like backdoors, and protecting against malware by catching threats early. However, it lacks verification for actual threats, leading to false positives, and may not detect serious or malicious activities if used improperly. Thus, while useful for preliminary checks, it should be complements

Commander Here is the final response:
The Pros and Cons of Using a Null Scan on a Windows Machine
A Null Scan on a Windows machine offers several benefits, including:

- **Efficiency:** Identifying potential vulnerabilities without executing harmful code can help prevent security breaches.
- **Smooth User Experience:** A Null Scan can provide a seamless user experience by detecting hidden risks like backdoors and protecting against malware.
- **Early Threat Detection:** Catching threats early can help prevent system compromise and data theft.

However, there are also some drawbacks to consider:

Latency:

Editor-28.34s

Researcher-164.58s

Manager-65.96s

Note: final response does not complete.

Final Response:

This screenshot shows the same interface as the previous one, but with the final response displayed. The 'Commander' message now includes the full response: 'The Pros and Cons of Using a Null Scan on a Windows Machine'. The response lists three pros: Efficiency, Smooth User Experience, and Early Threat Detection. It also lists a drawback: 'However, there are also some drawbacks to consider:'. The Manager's response is still partially visible at the bottom.

Cybersecurity Incident Response Agent

Explain the pros and cons of using a Null Scan on a windows machine

Here is the final response:
The Pros and Cons of Using a Null Scan on a Windows Machine
A Null Scan on a Windows machine offers several benefits, including:

- **Efficiency:** Identifying potential vulnerabilities without executing harmful code can help prevent security breaches.
- **Smooth User Experience:** A Null Scan can provide a seamless user experience by detecting hidden risks like backdoors and protecting against malware.
- **Early Threat Detection:** Catching threats early can help prevent system compromise and data theft.

However, there are also some drawbacks to consider:

Upon observation, the 2-bit model of llama 3:8b loses the logical 'thread' when compared to the 8-bit version of llama3:8b, especially when it comes to completing the task and maintaining coherence. The 2-bit version of the model hallucinated the pros of a Null Scan as well as failed to completely provide a con for a Null Scan. In the 8-bit version of the model, the final response was coherent as a Null Scan is indeed a useful tool for hackers and was able to provide 2 pros and almost completed a con. On the brighter side, at least it was able to maintain coherence.